

Lab 09 Template

Some questions require multiple parts to be answered, be sure to discuss them in full.

Part 01:

1. **Screenshot of the output from** `openssl ec -text -in <netid>_private_key.pem |less`
(5 points)
2. **What is the difference between the two curve identifiers?**
(5 points)
3. **What does the CONNECTED(00000003) line indicate about the steps that occurred before establishing the SSL/TLS handshake?**
(5 points)
4. **Take a screenshot of the client key_share information**
(5 points)
5. **What information do you think the client key_share fields are providing to the server? How might the server use this information during the TLS 1.3 handshake to establish a secure connection? Does this contribute to perfect forward secrecy? Why or why not?**
(10 points total, 2.5 points each question)
6. **Take a screenshot of the supported version extension for the server. What version(s) of TLS does the server actually support?**
(10 points total, 5 points screenshot, 5 points question)
7. **Take a screenshot of the key_share information.**
(5 points)
8. **Take a screenshot of the certificate chain**
(5 points)
9. **Look at the subject (s) and the issuer (i) for each of the two certificates. Which number is the certificate for the Certificate Authority (CA)? Why?**
(10 points total, 5 points for each question)
10. **What happens if the server's certificate validation fails at any depth?**
(5 points)

11. **Expand the acronym for each piece of the cipher suite and explain the purpose of each.**
(10 points total, 2 points each piece)
12. **What does “getting new key material” mean?**
(5 points)
13. **Why would creating pkcs12 files be a bad idea generally speaking?**
(5 points)
14. **Take a screenshot of the ciphers supported in s_server binary.**
(5 points)
15. **Find and record in your lab report at least one that should no longer be used in the real world. Support your reason why it shouldn't be used.**
(10 points total, 5 points each answer)